Similar Predictions, Different Weights: How AdamW and Muon Train a Small Language Model

Initial checkpoint, MIT 6.7910, Fall 2026

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Summary. We train a small language model with AdamW and with Muon, from one initialization and on the same mini-batches. (1) Muon trains faster and on a different path. At equal training loss, their predictions differ about as much as two runs on different data orders; their final weights differ more. (2) Muon's weight and update matrices have more evenly sized singular values than AdamW's, beyond data-order variation; no prediction statistic differs. (3) A run switched to the other optimizer mid-training ends up predicting no further from either optimizer's own run than two data orders do. Its updates take the new optimizer's form; its weights and activations keep part of the old one's.

Setup. *Model.* We train a 14M-parameter nanoGPT (Karpathy, 2022) on TinyStories (Eldan and Li, 2023) for 10,000 steps of 128 sequences of 256 tokens, with a cosine learning-rate decay (Appendix A).

Optimizers. Muon (Jordan et al., 2024) orthogonalizes the update of each weight matrix in the transformer blocks: it sets the update's singular values to about 1. AdamW trains Muon's other parameters. Each optimizer uses the best of five learning rates. All runs start from the same weights. A *data order* fixes the mini-batch of every step; each optimizer runs on five.

Comparing runs. The distance between two runs is the mean Jensen–Shannon divergence (JS, in bits) of their next-token predictions on held-out text. We compare runs at equal training loss, so that speed does not count as a difference. The *floor* is the mean distance between two data orders of one optimizer (for two optimizers, the larger floor). A distance of one floor means "as different as two data orders".

Ratios. For any other metric, we compare two optimizers by the ratio of their values at equal training loss. If their values range over $[a_{\min}, a_{\max}]$ and $[b_{\min}, b_{\max}]$ across data orders, the ratio lies in $[a_{\min}/b_{\max},\ a_{\max}/b_{\min}]$. We write it as midpoint ± half-width, the *margin*. If the interval excludes 1, the optimizers differ by more than data orders do.

Swaps. A swap switches a run of one optimizer (the *parent*, on data order 0) to the other at a *fork step*, then trains it to step 10,000 on the parent's batches (Appendix C). An optimizer's *own run* is its run from the start.

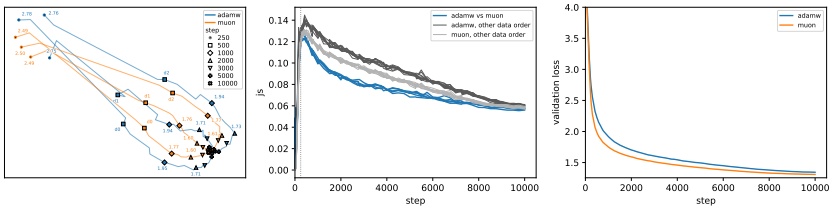


Figure 1: AdamW and Muon. (a) Checkpoints from step 250 on, mapped to 2-D by densMAP (shows neighbours, not distances); labels: training loss and data order. (b) JS between runs at equal step. (c) The same at equal training loss. (d) Validation loss.

Step 1: how the trajectories differ (Fig. 1). After the learning-rate warmup, AdamW and Muon runs on the same data order stay close: they are nearest neighbours in the densMAP plot (a). Muon learns faster: it reaches AdamW's final validation loss, 1.344, by step 7500 (d).

Muon does not just follow AdamW's path faster. The two share the initialization and the batches. If Muon only moved faster along AdamW's path, their distance at equal training loss would be far below the floor. Instead it sits at the floor (c): 0.95–1.38 floors below training loss 4.85 (median 1.02), and 0.99 at the lowest loss all runs reach. So switching the optimizer moves the predictions about as much as changing the data order.

If Muon were only faster, comparing at equal loss would also bring the two closer than comparing at equal step. The opposite happens: their JS is 0.102 and 0.079 at steps 1000 and 3000 (b), but 0.131 and 0.106 at AdamW's training losses at those steps (c). At equal step they are even closer than two data orders of AdamW (0.127 and 0.103), likely because both models have just trained on the same batches.

Their final weights are further apart than two data orders'. The *loss barrier* between two weight vectors is how far the training loss rises, on the straight path between them, above the line joining its end values (Frankle et al., 2020). It is 3.5–4.1 nats between AdamW and Muon, against 1.2–1.9 between two AdamW data orders and 1.6–1.8 between two Muon ones.

Step 2: what differs along the way. We measure 30 properties of each model at 34 points of training, in five groups: weights W, one-step updates ΔW, loss curvature (Hessian H), internal activations and predictions (Appendix B). We compare optimizers by each metric's ratio at training loss 1.81, with three data orders each (Fig. 2, Table 3). SGD with momentum, our reference, differs from both in most metrics (Table 4).

- *Weights W.* Muon's weight matrices use their directions more evenly. Their stable rank $\|W\|_F^2/\|W\|_2^2$ is 86 against AdamW's 33 (2.61 ± 0.04 times), and their effective rank 325 against 272. Their spectral norm (2.9 against 5.4) and largest-to-median singular-value ratio $\sigma_1/\sigma_{\text{median}}$ (2.7 against 8.0) are lower. Both move equally far from the initialization (ratio 1.01 ± 0.02): they differ in the shape of their weights, not in how far they travel.
- *One-step updates ΔW.* Consecutive Muon steps are less alike (cosine 0.74 against 0.88). Their stable rank is much higher (220 against 17), but by construction: Muon orthogonalizes every update.
- *Loss curvature (Hessian H).* Muon is about twice as sharp: the top Hessian eigenvalue $\lambda_{\max}$ is 14 against 6.2, the trace $\operatorname{tr} H$ 2770 against 1400. But the tuned learning rates differ, and $\lambda_{\max}$ scales roughly as one over the learning rate (log-log slopes -0.86 and -0.84 across a sweep). So part of this gap reflects the rates, not the optimizer.
- *Internal activations.* AdamW's MLP is sparser: 12% of its units are active, against 22% for Muon.
- *Predictions.* No prediction metric separates the two: every AdamW/Muon interval contains 1.

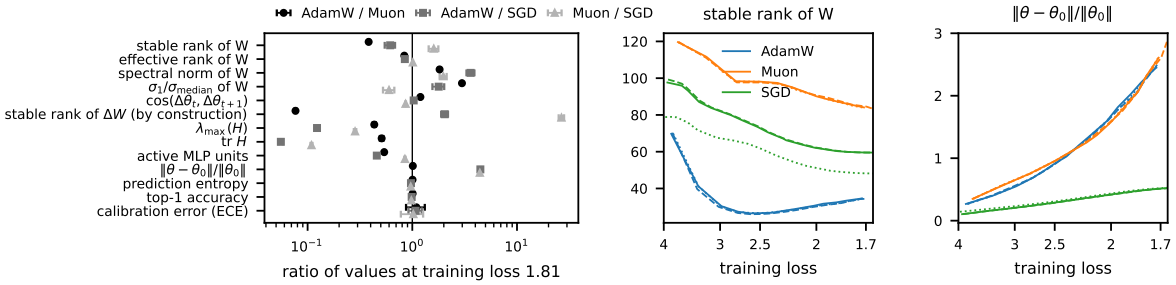


Figure 2: Three data orders per optimizer. (a) Ratio of two optimizers' values at training loss 1.81, for selected metrics; bars: margins. (b), (c) Stable rank of W and distance from initialization against training loss.

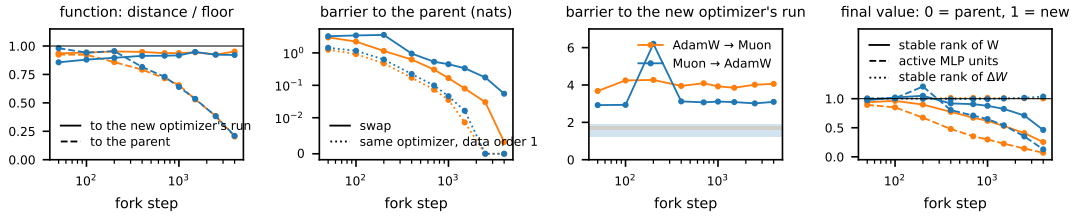


Figure 3: Swaps: the swapped run's final model against the fork step. (a) JS to the new optimizer's own run and to the parent, in floors. (b) Loss barrier to the parent. (c) Loss barrier to the new optimizer's own run; band: between two of its data orders. (d) Three metrics on a scale from the parent's value (0) to the own run's (1).

Step 3: swaps (Fig. 3). We swap at nine fork steps from 50 to 4000, in both directions (Appendix C). In its predictions, every swapped run ends as similar to both its parent and the new optimizer's own run as two data orders are: 0.86–0.95 floors from the own run, and 0.92–0.98 floors from the parent at fork step 50 (0.20–0.21 at 4000).

Its weights are not. It is *linearly connected* to its parent (barrier below 0.05) only from fork step 2500 when leaving AdamW, and never when leaving Muon. Keeping the optimizer and switching only the data order gives this from fork step 1000. The swapped run also stays further from the new optimizer's own run than that optimizer's data orders are from each other (c).

In the final model (d), the updates take the new optimizer's form at every fork step: the stable rank of ΔW moves 0.99–1.04 of the way from the parent's value to the new optimizer's. The weights and activations keep part of the parent's, more so the later the swap. At fork step 4000, the stable rank of W moves 0.25 (AdamW to Muon) and 0.46 (Muon to AdamW) of the way, and the fraction of active MLP units 0.07 and 0.13.

Next steps. (i) Repeat Steps 1–3 on a larger model (TinyStories-30M) and on character-level Shakespeare. (ii) For each property that differs in Step 2: does it follow directly from Muon's orthogonalized updates, or does it emerge during training? Does the sharpness gap come from the different tuned learning rates? (iii) Do these differences explain Step 1, Muon's faster training and different path? Is any of them a reason Muon does better? (iv) Talk (Dec 1–4); paper, code, Colab (Dec 5–11). https://github.com/qqgjyx/MIT-6.7910

References

Eldan, R. and Y. Li (2023). *TinyStories: How Small Can Language Models Be and Still Speak Coherent English?*

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A Hyperparameters

Table 1: Hyperparameters shared by the AdamW and Muon runs.

Runs	AdamW and Muon × data orders d0–d4 (10)
Model	nanoGPT, pre-LayerNorm blocks
Layers, heads, width	6, 6, 384
MLP	GELU, width 1536
Biases, dropout	none, none
Output head	untied
Attention projections	separate query, key, value
Context	256 tokens
Parameters	13.9M (10.6M without token and position embeddings and head)
Data	TinyStories (GPT-4 version)
Tokenizer	byte-level BPE, 4096 tokens, trained on the training split
Batch	128 windows × 256 tokens
Window offsets	from the training split; generator seeded with (data order, step), same for every optimizer
Steps	10,000
Tokens	328M (about 0.6 epochs)
Learning rate	$\eta\,\varphi(t)$ at step t, every parameter group
Warmup	linear, $\varphi(t) = (t+1)/200$ for $t < 200$
Decay	cosine to $\varphi = 0.1$ at step 10,000
Gradient clipping	none
Initialization seed	0, all runs
Training precision	bfloat16 autocast
Prediction, Hessian precision	float32, no autocast (TF32 matrix products allowed)
Logged predictions	next-token distributions: 128 fixed validation sequences × 16 evenly spaced positions (2,048), as float16 log-probabilities
Training loss	64 fixed training sequences, all positions; used to match runs at equal loss
Both logged at	every 250th step + 30 log-spaced steps in $[1, 10{,}000]$
Hessian	top eigenvalue, 20 Lanczos iterations, 32 fixed training sequences; AdamW also of $P^{-1/2}HP^{-1/2}$, with P AdamW's diagonal preconditioner
Measured at	every 500th step + 30 log-spaced steps
Loss barrier	64 fixed training sequences, 11 evenly spaced points of the segment between the final weights

Table 2: Optimizer-specific hyperparameters. n_{out}, n_{in}: output and input size of the matrix.

	AdamW	Muon
η	$2 \cdot 10^{-3}$	0.01
η choice	best final validation loss of a 5-rate sweep on d0	as AdamW
Update	bias-corrected moments	Nesterov momentum; 5 quintic Newton–Schulz steps in bfloat16
Momentum	$\beta = (0.9, 0.95)$	$\beta = 0.95$
ϵ	10^{-8}	–
Update scale	–	$\sqrt{\max(1, n_{\text{out}}/n_{\text{in}})}$
Weight decay	0.1, decoupled; none on LayerNorm gains	none
Parameters	all	query, key, value, attention-output and both MLP matrices of every block
Aux. AdamW	–	embeddings, head, LayerNorm gains
Aux. η, β, ϵ	–	$2 \cdot 10^{-3}$, $(0.9, 0.95)$, 10^{-8}
Aux. weight decay	–	none

B Metrics

The Step-2 runs save checkpoints every 1000 steps, at 12 log-spaced steps, and at three consecutive steps $t, t+1, t+2$ for 24 values of t from 25 to 9998 (to measure one-step updates); we measure the metrics at 34 of these per run, and the Step-3 swaps only at their final weights. *Weights*: for all parameters together, $\|\theta\|_2$, $\|\theta - \theta_0\|/\|\theta_0\|$ and $\|\theta\|_\infty/\|\theta\|_2$; and, averaged over the 36 weight matrices W of the blocks, the stable rank $\|W\|_F^2/\|W\|_2^2$, the effective rank $\exp H(\sigma/\sum\sigma)$ with H the entropy of the normalized singular values σ, the spectral norm, and $\sigma_1/\sigma_{\text{median}}$. *Updates* $\Delta\theta_t = \theta_{t+1} - \theta_t$: $\|\Delta\theta_t\|/\|\theta_t\|$, its cosine with $\Delta\theta_{t+1}$ and with the negative gradient $-g_t$ on step t's batch, and the stable and effective rank of its block matrices ΔW. *Curvature* of the training loss on 32 fixed training sequences, with Hessian H: $\lambda_{\max}(H)$

(20 Lanczos iterations, logged during training), $\operatorname{tr} H$ (10 Rademacher probes), $\operatorname{tr} H/\lambda_{\max}$, and the batch sharpness $\mathbb{E}_B[g_B^\top H_B g_B]/\mathbb{E}_B\|g_B\|^2$, where g_B and H_B are the gradient and Hessian on a mini-batch B. The log-log slopes of $\lambda_{\max}$ against the learning rate in Step 2 are fitted across a learning-rate sweep, against the peak rate of the schedule. *Predictions*, on the 2,048 held-out positions: entropy, mean top-1 probability, calibration error (15 bins), top-1 accuracy, loss on the targets in the top and the bottom third of training-set frequency and their difference, and JS to the unigram and to the bigram model of the training split. *Activations*, on 32 held-out sequences: attention entropy and attention on the first token (means over layers, heads and positions), fraction of MLP pre-activations above 0, RMS of the residual stream after the last block, and effective rank of the centred final features.

SGD, the reference in Table 4, uses heavy-ball momentum 0.9 and learning rate 0.06, the only rate tried between 0.03 and 0.12 that did not diverge on any of the three data orders; everything else is as in Table 1.

Table 3: All metrics of Step 2 at training loss 1.81: mean ± half-range over data orders 0–2, and the ratio AdamW/Muon with its margin. Rows marked ($\times 10^k$) are in units of 10^k. Bold: the five ratios farthest from 1 whose interval excludes 1.

	AdamW	Muon	A/M
Weights W			
weight norm $\|\theta\|_2$	224 ± 2	224 ± 0	0.997 ± 0.013
distance from init. $/\|\theta_0\|$	2.18 ± 0.03	2.15 ± 0.01	1.01 ± 0.02
largest entry $/\|\theta\|_2$ ($\times 10^{-3}$)	7.04 ± 0.06	6.45 ± 0.02	1.09 ± 0.01
stable rank of W	33.0 ± 0.3	86.2 ± 0.3	$\mathbf{0.383 \pm 0.005}$
effective rank of W	272 ± 0	325 ± 0	0.839 ± 0.000
spectral norm of W	5.39 ± 0.07	2.95 ± 0.01	1.82 ± 0.03
$\sigma_1/\sigma_{\text{median}}$ of W	8.01 ± 0.05	2.68 ± 0.00	$\mathbf{2.98 \pm 0.03}$
One-step updates ΔW			
step size $\|\Delta\theta\|/\|\theta\|$ ($\times 10^{-3}$)	6.56 ± 0.08	6.53 ± 0.03	1.01 ± 0.02
stable rank of ΔW	16.7 ± 0.3	220 ± 0	$\mathbf{(7.66 \pm 0.14) \times 10^{-2}}$
effective rank of ΔW	228 ± 1	369 ± 0	0.618 ± 0.003
cosine of consecutive steps	0.883 ± 0.002	0.738 ± 0.003	1.19 ± 0.01
cosine of step and $-g$	0.244 ± 0.004	0.217 ± 0.007	1.12 ± 0.06
Loss curvature (Hessian H)			
top Hessian eigenvalue	6.18 ± 0.10	14.3 ± 0.2	$\mathbf{0.433 \pm 0.013}$
batch sharpness	0.974 ± 0.254	3.13 ± 0.37	$\mathbf{0.332 \pm 0.116}$
Hessian trace	1400 ± 40	2770 ± 20	0.510 ± 0.019
trace / top eigenvalue	224 ± 6	196 ± 7	1.16 ± 0.08
Internal activations			
attention entropy (nats)	2.51 ± 0.03	2.79 ± 0.02	0.899 ± 0.014
attention on first token ($\times 10^{-2}$)	1.75 ± 0.11	1.98 ± 0.05	0.908 ± 0.079
active MLP units	0.120 ± 0.001	0.222 ± 0.001	0.539 ± 0.007
residual-stream RMS	4.73 ± 0.05	3.38 ± 0.02	1.40 ± 0.03
effective rank of features	312 ± 2	320 ± 0	0.977 ± 0.007
Predictions			
prediction entropy (nats)	1.89 ± 0.01	1.88 ± 0.01	1.01 ± 0.01
mean top-1 probability	0.561 ± 0.003	0.563 ± 0.001	0.994 ± 0.007
calibration error ($\times 10^{-2}$)	3.25 ± 0.17	3.05 ± 0.46	1.09 ± 0.22
top-1 accuracy	0.546 ± 0.005	0.544 ± 0.005	1.01 ± 0.02
loss on frequent targets	0.896 ± 0.004	0.892 ± 0.008	1.00 ± 0.01
loss on rare targets	3.10 ± 0.02	3.10 ± 0.01	1.00 ± 0.01
rare minus frequent loss	2.21 ± 0.02	2.21 ± 0.01	0.996 ± 0.015
JS to unigram model	0.800 ± 0.001	0.802 ± 0.001	0.998 ± 0.002
JS to bigram model	0.469 ± 0.001	0.471 ± 0.001	0.996 ± 0.004

Table 4: The metrics of Table 3 with SGD as a reference: ratios AdamW/SGD and Muon/SGD. Bold: in each ratio column, the five ratios farthest from 1 whose interval excludes 1.

	AdamW	Muon	SGD	A/S	M/S
Weights W					
weight norm $\|\theta\|_2$	224 ± 2	224 ± 0	110 ± 0	2.04 ± 0.02	2.05 ± 0.01
distance from init. $/\|\theta_0\|$	2.18 ± 0.03	2.15 ± 0.01	0.486 ± 0.006	4.47 ± 0.11	4.42 ± 0.07
largest entry $/\|\theta\|_2$ ($\times 10^{-3}$)	7.04 ± 0.06	6.45 ± 0.02	39.5 ± 1.0	0.180 ± 0.006	$\mathbf{0.165 \pm 0.005}$
stable rank of W	33.0 ± 0.3	86.2 ± 0.3	56.2 ± 5.6	0.614 ± 0.069	1.60 ± 0.17
effective rank of W	272 ± 0	325 ± 0	321 ± 1	0.849 ± 0.003	1.01 ± 0.00
spectral norm of W	5.39 ± 0.07	2.95 ± 0.01	1.47 ± 0.11	3.61 ± 0.31	1.98 ± 0.15
$\sigma_1/\sigma_{\text{median}}$ of W	8.01 ± 0.05	2.68 ± 0.00	4.37 ± 0.55	1.79 ± 0.23	0.599 ± 0.074
One-step updates ΔW					
step size $\|\Delta\theta\|/\|\theta\|$ ($\times 10^{-3}$)	6.56 ± 0.08	6.53 ± 0.03	0.624 ± 0.047	$\mathbf{10.9 \pm 1.0}$	$\mathbf{10.8 \pm 0.9}$
stable rank of ΔW	16.7 ± 0.3	220 ± 0	8.37 ± 0.48	2.04 ± 0.15	$\mathbf{26.6 \pm 1.6}$
effective rank of ΔW	228 ± 1	369 ± 0	205 ± 2	1.12 ± 0.02	1.81 ± 0.02
cosine of consecutive steps	0.883 ± 0.002	0.738 ± 0.003	0.857 ± 0.004	1.03 ± 0.01	0.863 ± 0.007
cosine of step and $-g$	0.244 ± 0.004	0.217 ± 0.007	0.447 ± 0.010	0.550 ± 0.022	0.491 ± 0.027
Loss curvature (Hessian H)					
top Hessian eigenvalue	6.18 ± 0.10	14.3 ± 0.2	50.2 ± 0.6	$\mathbf{0.123 \pm 0.004}$	0.284 ± 0.008
batch sharpness	0.974 ± 0.254	3.13 ± 0.37	6.38 ± 0.50	$\mathbf{0.162 \pm 0.051}$	0.504 ± 0.095
Hessian trace	1400 ± 40	2770 ± 20	$(2.55 \pm 0.04) \times 10^4$	$\mathbf{(5.55 \pm 0.25) \times 10^{-2}}$	$\mathbf{0.109 \pm 0.002}$
trace / top eigenvalue	224 ± 6	196 ± 7	507 ± 9	0.447 ± 0.020	0.385 ± 0.021
Internal activations					
attention entropy (nats)	2.51 ± 0.03	2.79 ± 0.02	3.84 ± 0.01	0.655 ± 0.009	0.728 ± 0.007
attention on first token ($\times 10^{-2}$)	1.75 ± 0.11	1.98 ± 0.05	1.82 ± 0.07	0.972 ± 0.095	1.07 ± 0.07
active MLP units	0.120 ± 0.001	0.222 ± 0.001	0.262 ± 0.002	0.458 ± 0.008	0.850 ± 0.012
residual-stream RMS	4.73 ± 0.05	3.38 ± 0.02	0.244 ± 0.003	$\mathbf{19.4 \pm 0.5}$	$\mathbf{13.8 \pm 0.3}$
effective rank of features	312 ± 2	320 ± 0	297 ± 2	1.05 ± 0.01	1.07 ± 0.01
Predictions					
prediction entropy (nats)	1.89 ± 0.01	1.88 ± 0.01	1.94 ± 0.00	0.971 ± 0.006	0.966 ± 0.007
mean top-1 probability	0.561 ± 0.003	0.563 ± 0.001	0.558 ± 0.001	1.00 ± 0.01	1.01 ± 0.00
calibration error ($\times 10^{-2}$)	3.25 ± 0.17	3.05 ± 0.46	2.97 ± 0.28	1.08 ± 0.16	1.02 ± 0.25
top-1 accuracy	0.546 ± 0.005	0.544 ± 0.005	0.554 ± 0.003	0.987 ± 0.013	0.980 ± 0.014
loss on frequent targets	0.896 ± 0.004	0.892 ± 0.008	0.851 ± 0.012	1.05 ± 0.02	1.05 ± 0.02
loss on rare targets	3.10 ± 0.02	3.10 ± 0.01	3.11 ± 0.04	0.999 ± 0.018	0.999 ± 0.015
rare minus frequent loss	2.21 ± 0.02	2.21 ± 0.01	2.26 ± 0.03	0.979 ± 0.020	0.982 ± 0.017
JS to unigram model	0.800 ± 0.001	0.802 ± 0.001	0.804 ± 0.001	0.995 ± 0.002	0.997 ± 0.003
JS to bigram model	0.469 ± 0.001	0.471 ± 0.001	0.462 ± 0.001	1.02 ± 0.00	1.02 ± 0.00

C Swaps

Table 5: Swaps. A: parent's optimizer; B: optimizer after the swap; s: fork step.

Parents	the d0 runs of AdamW and Muon (Table 1)
Warm state	the other optimizer, stepped on the same mini-batch gradients with learning rate 0: momenta, second moments and step counts evolve as if training; parameters follow A alone
Fork steps s	50, 100, 200, 400, 700, 1000, 1500, 2500, 4000
Saved at s	parameters; states of both optimizers
Swap $A \to B$ loads	parameters at s; B's warm state
Hyperparameters	B's (Table 2)
Steps	s to 10,000, on the parent's schedule and batches (data order 0)
Swap forks	2 directions $\times$ 9 steps = 18
Same-batch control	the parent itself (A continued with the same batches)
Data-order control	A continued from s with its own state, data order 1: 18 forks